

5. Humans are protected from the effects of disease by vaccination programmes and the use of antibiotics.

(a) (i) Tick (✓) **three** boxes next to the correct statements about vaccines. [3]

- vaccines contain antigens

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- vaccines contain lymphocytes

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- vaccines stimulate the production of antibodies

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- vaccines only protect against viruses

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- vaccines can contain modified viruses

☐
- vaccines contain antibodies

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(ii) Measles is a highly infectious viral illness that is uncommon in the UK because of the effectiveness of vaccines. The measles vaccine provides life-long protection against the disease.

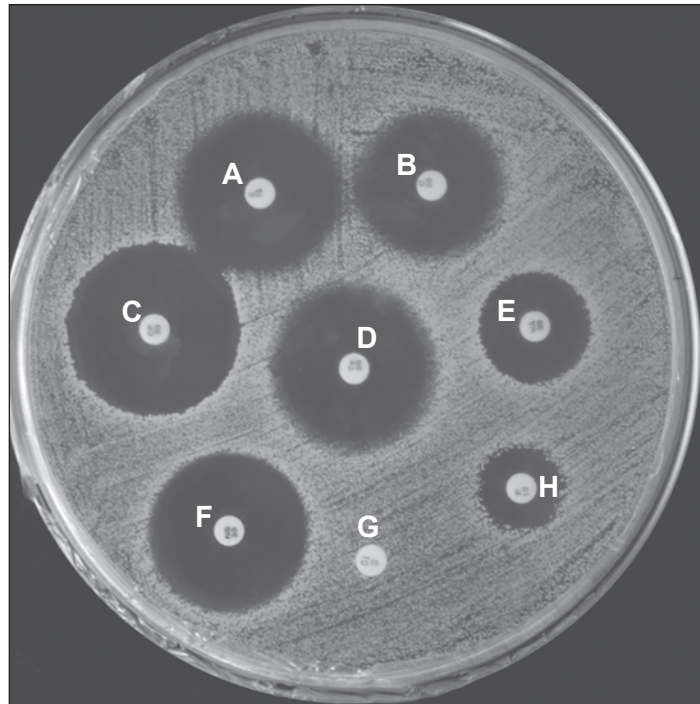
Flu vaccines are also available. Unlike the measles vaccine, they need to be given every year.

State **one** reason why a flu vaccine needs to be given every year. [1]

- (b) An antibiotic is a substance that slows down or stops bacterial growth. Not all antibiotics work against all bacteria. Therefore it is important to know which antibiotics are most effective against different bacteria.

The photograph shows the results from an experiment to determine the effectiveness of different antibiotics against one type of bacteria.

Eight different antibiotics were tested. Each disc contains one of the antibiotics (**A** to **H**).



A website states that not all antibiotics are effective against this type of bacteria. With reference to **all** antibiotics (**A-H**), explain whether you agree with the website. [3]

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